Supplementary File S7: November 1–3 Simulator Evaluation Evidence

November 3, 2026

Scope and evidence selection

This file documents the simulator evidence cited in the revised evaluation. It keeps the integrated pipeline, scene–protocol probes, five-seed policy comparison, process-window ablation, approval pilot, production-mode checks, and the negative live-Agent pilot separate because they use different test units and denominators. The raw CSVs, run manifests, reports, and the self-contained visual report are in `Supplementary-File-S7-data-2026-11-02.zip`. No physical-plant result is included.

Consolidation update (November 2–3, 2026)

The data package is organized into three folders. `01-paper-evidence/` is the frozen November 1 batch used for every positive number in the manuscript. `02-c3-pilot-negative/` contains a November 2 live-Agent multi-round pilot. The manuscript reports this pilot as a negative result in Section VI-G; no C3 number is used for a positive claim. After pipeline fixes, the best episode completed with zero governed writes and a process variable outside its band, and a second episode ended at its time budget with zero writes. The pilot exposed platform integration defects (task-artifact addressing, lead queue draining, and goal completion without checking the goal criterion) that remain outside the paper's positive claims. `03-system-validation-not-used/` contains integration and demonstration reports that validate the platform but are not part of the controlled experiment blocks.

A cast-film probe record from run `20261101121700-scenario-probes` was excluded. Its manifest declared four scenes but recorded only cast-film, and its simulator endpoint did not match the endpoint in the reproduction command. None of its 24 admissible requests, 24 structural probes, or 20 readbacks are included in the manuscript totals or the data package. The reported scene–protocol counts below use only run `20261101121221-scenario-probes`.

Representative systems

Table S7-I summarizes the operation boundary and evaluation scope of representative plant-facing agent systems cited in the manuscript. It does not rank their performance, and the reported evaluations are not directly comparable because the systems, tasks, and denominators differ.

Integrated pipeline and scene–protocol probes

The integrated executions `20261101103641-9j8` and `20261101104849-xu0` used seed 42 and reported commit `da627d21`. Both manifests record that the platform and simulator were already running and that no Agent harness was active. Each run returned 77 passing assertions, zero warnings, and zero failures. Assertions are within-run checks, not independent trial observations.

TCP and RTU in the table denote Modbus TCP and Modbus RTU. For each writable node, four tested values were expected to pass (three interior values and the upper boundary), followed by four structural-range probes expected to reject. One structural case in the biaxial-line set was not collected. Readback is counted only when the driver returned an actuator value. Across this four-scene subset, the totals are 220/220 accepted admissible requests, 219/219 rejected structural probes, and 216/216 matching available readbacks. The full planned scene–protocol matrix remains incomplete.

The integrated film-line preset check is a separate pipeline result: five endpoint types were configured and nine selected negative probes were rejected (9/9), with no code edits. It reused the same runtime and is not an independent cold-start deployment. The pipeline run manifests and summaries are included in the data archive.

Table 1: Table S7-I. Representative plant-facing agent systems and the scope of their reported evaluations.

System	Reported operation or integration mechanism	Reported evaluation
Biskupovic et al.	Agent-based service selection and online reselection using performance and computational-load metrics	Industrial cyber–physical system study; service adaptability and performance
Köhle et al.	Module-specific agents, MCP-to-OPC-UA skills, MQTT coordination, and injected factory state	Nine simulated production challenges
Xavier et al.	MCP adapters for industrial protocols with a common response envelope and mock-first testing	870 benchmark runs and 2,820 tool calls
Graul et al.	AAS-grounded proposals, signed approval, PLC-validated OPC UA Methods, and audit log	Two laboratory proof-of-concept settings
Schall	P&ID-, SCADA-, and SOP-grounded proposal verification with gated execution	Five simulated scenarios, 43 proposals, and 50 LLM runs
Lin and Chen	Policy-governed sensor–MES–ERP digital thread	Manufacturing-record evaluation with operational metrics
This work	Scene-bound production context and named signal mappings; shared dispatch with route-specific write governance and separate command receipts and process-observation evidence	Two pipeline runs (77 assertions each), a four-scene probe subset, five-seed scripted missions, a three-repeat mock-driver ablation, and a negative live-Agent pilot

Table 2: Valid collected scene–protocol probe outcomes.

Scene	Writable nodes	Writable nodes by protocol	Admissible accepted	Structural rejected	Readback matched
Injection	11	TCP 5; RTU 2; OPC UA 3; MQTT 1	44/44	44/44	44/44
Wastewater	7	TCP 1; RTU 3; OPC UA 2; MQTT 1	28/28	28/28	28/28
Annealing	7	TCP 3; RTU 2; OPC UA 1; MQTT 1	28/28	28/28	28/28
Biaxial line	30	TCP 20; RTU 1; OPC UA 8; HTTP 1	120/120	119/119	116/116

Five-seed target-band comparison

C0 is a 60-s no-write condition; its terminal process-variable statistic is the mean of the final five samples. C1 is a deterministic scripted policy that calls the `recipe_trial` batch route of the shared governed-write service and uses scenario-specific mission logic; its statistic is the first in-band reading at which the mission stopped, not a fixed-horizon endpoint. Each seed appears in both conditions for injection mass, wastewater dissolved oxygen (DO), and annealing hardness. C1 per-seed wall time was 150.41–172.90 s, compared with C0's fixed 60 s, so this is a descriptive contrast rather than a time-matched test. One C0 control, seed 42, began from residual state (injection 32.378 g and DO 2.894 mg/L, both inside their bands) rather than the blueprint state used by C1; seeds 43–46 began from the blueprint. Counts are reported for the 12 blueprint-matched pairs and separately for all 15 runs. Counts are recomputed from raw `pv`, `target`, and `tol` fields. The runner's `attained` flag has different condition-specific semantics (C0 combines the PV band with configured guards; C1 records its scenario mission result), so it is retained as source data but not compared across conditions.

Table 3: Raw endpoints with a common target-band check recomputed from each process variable, target, and tolerance: 0/15 C0 and 15/15 C1 endpoints were in band overall, and 0/12 C0 and 12/12 C1 for the blueprint-matched seeds 43–46. The condition-specific runner flags are not used for this count.

Scene	Target band	Seed	C0 endpoint	C1 endpoint	PV band C0/C1
Injection	32.5 ± 0.35 g	42	31.226	32.320	No / Yes
Injection	32.5 ± 0.35 g	43	31.272	32.180	No / Yes
Injection	32.5 ± 0.35 g	44	31.244	32.173	No / Yes
Injection	32.5 ± 0.35 g	45	31.272	32.190	No / Yes
Injection	32.5 ± 0.35 g	46	31.206	32.340	No / Yes

Scene	Target band	Seed	C0 endpoint	C1 endpoint	PV band C0/C1
Wastewater DO	3.4 ± 0.6 mg/L	42	0.202	3.864	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	43	0.198	3.800	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	44	0.268	3.686	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	45	0.200	3.576	No / Yes
Wastewater DO	3.4 ± 0.6 mg/L	46	0.226	3.646	No / Yes
Annealing	95 ± 6 HV	42	134.66	97.74	No / Yes
Annealing	95 ± 6 HV	43	134.60	97.44	No / Yes
Annealing	95 ± 6 HV	44	134.66	97.80	No / Yes
Annealing	95 ± 6 HV	45	134.90	99.90	No / Yes
Annealing	95 ± 6 HV	46	134.36	98.20	No / Yes

The individual values above are copied from the ten per-seed CSV result files. Across the 12 blueprint-matched pairs, 0/12 C0 and 12/12 C1 endpoints were in band; including the warm-started seed-42 control gives 0/15 and 15/15. Across the five C1 seeds, the scripted missions issued 12 writes in 17 rounds for injection, 56 writes in 59 rounds for wastewater, and 41 writes in 50 rounds for annealing; these are workload counts, not an efficiency comparison. Seed values are simulator initializations, not independent production lines. C1 is not an active-LLM policy and C0 is not an industrial-controller baseline.

Process-window guard ablation

The process-window study used seed 42, three repetitions in each arm, a direct DCW API call, a mock driver, and benchmark mode. The configuration hash was `3a57e722d4764b98`. Each repetition submitted six negative probes: four hard structural probes and two process-window probes; it also submitted three nominal legal requests and three boundary requests. The additional 20 timed writes per repetition were used for latency summaries and were excluded from these denominators. No readback-mismatch stimulus was included.

Table 4: Three-repeat process-window ablation. The combined-removal arm removes the process-window and readback-mismatch checks; hard structural limits remain active in every arm.

Arm	Negative probes rejected	Window violations reaching driver	Legal false blocks	Boundary expectations met
Full guard	18/18	0/6	0/9	9/9
No process-window guard (`no-interlock` arm)	12/18	6/6	0/9	9/9
No-readback check	18/18	0/6	0/9	9/9
No window + no-readback	12/18	6/6	0/9	9/9

The four hard structural probe types were blocked in every arm (12/12 per arm). With the process-window guard active, the two targeted window violations in each repetition did not reach the driver; removing that guard allowed all six across three repetitions to reach the mock driver. The no-readback arm does not test readback-mismatch rejection because that stimulus was absent. Benchmark mode relaxes normal timing and hold-window controls, so these results concern the tested guard path only.

Approval, production-mode, and authorization checks

The approval pilot used one injection scene and a scripted decision fixture. The full CSV and timing records are in the data archive.

Table 5: Four calls in the approval-route pilot. Auto rows are not clean matched pairs with the manual rows; no operator made a decision.

Mode	Runner stimulus	Decision	Before	After	Delay (ms)
Manual	legal-approved	approved	202	204.8	10 593
Manual	legal-rejected	rejected	216	216	544
Auto	label: manual arm rejected	n/a	202	204.8	10 071
Auto	label: manual arm approved	n/a	216	218.8	10 073

The manual reject did not change the simulated control value. The auto-mode label/value mismatch means the four calls do not form a clean paired approval-on/off experiment. This pilot demonstrates route handling only; it is not evidence of operator judgment or an isolated approval-mechanism effect.

In the separate production-mode guard run, the baseline value 190 was accepted. One example each was rejected for interval, step, node-range, recipe-window, and hold-window constraints. The exact API rejection codes are preserved in the source report and CSV. These one-case checks confirm expected branches for their configured inputs; they are not estimates of general rejection rates. A separate three-case authorization probe rejected an unbound write and a cross-node write while accepting one bound control write.

Reproduction and provenance

The platform manifests record source commit `da627d21`. The experiment runners were uncommitted at execution time and were later committed as `40533741`; the recorded commit therefore identifies the platform source, not a hermetic tree. The raw source records in the archive preserve runner outputs and exact run identifiers. Representative commands recorded by the runners are:

```
node bench/pipeline.mjs -profile integrated -seed 42 -cl-seeds 3

node bench/experiment/scenario-protocol-probes.mjs -scenarios injection,wwtp,anneal,biax -seed 42

node bench/experiment/optimization-control.mjs -condition C1 -scenarios injection,wwtp,anneal -seed 42

node bench/experiment/optimization-control.mjs -condition C0 -scenarios injection,wwtp,anneal -seed 42
-duration 60

node bench/e1-lite.mjs -seed 42 -repeats 3

node bench/experiment/approval-gate.mjs -scenario injection -base http://127.0.0.1:3005

node bench/experiment/production-guards.mjs -base http://127.0.0.1:3010

node bench/experiment/authorization-probe.mjs -scenario injection -base http://127.0.0.1:3005
```

The optimization commands were run for seeds 42–46 in both conditions. Re-running them creates new result identifiers and can produce different simulator trajectories if the environment or service state changes. The integrated pipeline reported a commit and harness hash, but the active service source tree was not captured as a hermetic snapshot. The archive includes the run manifests, selected reports, raw CSVs, a machine-readable validation summary, checksum manifest, and self-contained HTML dashboard; it does not include the excluded cast-film run.